When On-Chain Flow Signals Work and When They Don't: AMM-Layer Stress Propagation Across Five Stablecoin Episodes

A Provenance-Gated Analysis Using Curve Finance TokenExchange Logs

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Abstract

Empirical contagion studies routinely report cross-market linkage during crises, but rarely separate genuine *contagion* (a shift in linkage caused by the crisis) from constant *interdependence*. Using Tier-A on-chain Curve `TokenExchange` flow, we apply the Forbes–Rigobon regime test to stablecoin stress and find regime-switching contagion that is sharp and mechanism-specific. For the June 2023 USDT/Curve episode — an *endogenous* pool-imbalance shock — cross-pool flow coupling is statistically indistinguishable from zero in calm conditions ($\hat{\rho} = 0.09$, $n = 95$) and activates during the acute window ($\hat{\rho} = 0.53$, $n = 53$, $p < 10^{-4}$); the calm-to-panic shift is significant (Fisher $z = +2.82$, $p = 0.005$). The coupling peaks at lag 0, consistent with a common-shock mechanism rather than sequential transmission. Critically, none of the four *exogenous* episodes (Terra/LUNA, USDC/SVB, FTX, BUSD) shows this activation — even though all now have genuine Tier-A A/A pool pairs to test — so the null is a mechanism finding, not a data gap. We further show two positive structural results. First, on-chain arbitrage is *stabilizing* in calm markets (flow intensity negatively correlated with price deviation) and acute stress significantly bends this in three of five events, flipping it to *amplifying* where arbitrage capacity is overwhelmed (USDT/Curve, BUSD: Fisher $z > 3.6$) and strengthening it where on-chain liquidity absorbs an off-chain run (FTX: $z = -2.8$). Second, price discovery is on-chain for DeFi-native stress: the Curve pool price leads Binance (+1h $r = 0.28$, $p < 10^{-4}$) and deviates 37× more (12.5% vs 0.3%) for USDT/Curve, while the CEX leads only for the exogenous SVB bank run. Finally, on the machine-learning question, we run one unsupervised detector causally on each evidence layer: an online Hidden Markov Model on Tier-A on-chain state recovers endogenous DeFi-native stress that the Tier-B market layer registers at chance — for Terra/LUNA, causal AUROC 0.95 versus 0.50, raising the alarm 116 hours before the CEX basis — whereas market data is needed for the exchange- and regulation-borne shocks the pool never sees. Detection, not prediction, is the right frame: supervised cross-event prediction collapses under concept shift, and the two layers are complementary by mechanism. The broad implication: DeFi stress surveillance needs *separate monitoring architectures* for endogenous pool-imbalance events versus exchange-originating or regulatory shocks. A provenance-aware three-gate pipeline underpins every claim, blocking fixture and Tier-B data from reaching paper-level conclusions.

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CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → *Machine learning*.

Keywords

stablecoin; AMM; Curve Finance; DeFi stress; provenance; claim gate; contagion networks; online detection; hidden Markov model

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1 Introduction

Between May 2022 and March 2023, four distinct stablecoin systems experienced significant stress: the algorithmic UST/Terra collapsed; FTX's implosion triggered exchange runs; USDC briefly de-pegged following Silicon Valley Bank's failure; and Curve Finance pools became severely imbalanced in June 2023. Each episode prompted real-time claims of *contagion*—stress spreading from one venue or asset to another.

Most empirical analyses of these events rely on price data alone and treat all data sources as equally reliable. This paper addresses both limitations by introducing a provenance-aware framework that makes data quality an explicit input to every empirical claim, and then holding every result—including a supervised prediction benchmark—to the same evidentiary gate. That discipline is what lets us separate the findings that survive from the one that does not: the same gate certifies three structural results on Tier-A on-chain flow and rejects a predictive lift that fails to replicate out-of-event.

The surveillance gap. A price-based DeFi stress monitor observes only stablecoin basis widening—an aggregate market outcome—and is blind to the flow-level mechanism beneath it. Curve Finance `TokenExchange` logs, by contrast, record every swap with an exact block timestamp and exact token amount, freely available and updated every 12 seconds. This is a direct observation of the trading flow that *produces* the price move, not a derived signal. Our analysis shows that the cross-pool flow coupling during the June 2023 USDT/Curve episode is strong but *contemporaneous* with the price stress rather than leading it (Sections 5 and 8); the contribution is therefore a mechanism-level decomposition of how stress moves through the AMM layer, not an early-warning claim. No existing stablecoin contagion study uses TokenExchange flow as its primary evidence layer.

Contributions. *Methodologically*, we introduce a provenance-aware three-gate pipeline (data-quality tier → feature tier → statistical significance) and a claim-level taxonomy (A/A DEX-flow, A/A settlement, A/B directional, B/B contextual) that caps every cross-venue claim at its lowest tier and blocks fixture data from reaching any conclusion. *Empirically*, on five stress episodes we report four results: (1) regime-switching contagion — USDT/Curve cross-pool coupling rises from $\hat{\rho} = 0.09$ (calm) to 0.53 (panic), Fisher $z = 2.82$, $p = 0.005$ — present only for the endogenous shock, despite all five events carrying genuine Tier-A A/A pairs; (2) arbitrage flipping from stabilizing to amplifying under stress ($z = +3.84$ USDT/Curve, +3.69 BUSD, −2.80 FTX); (3) on-chain price discovery, the Curve pool leading the exchange and deviating 37× more for DeFi-native stress; and (4) a machine-learning result on which evidence layer detects stress first — an unsupervised online detector run causally on Tier-A on-chain state recovers endogenous DeFi-native stress that the Tier-B market layer registers at chance (Terra/LUNA causal AUROC 0.95 vs 0.50, 116h earlier), while market data is required for exchange-borne shocks; the layers are complementary, and supervised cross-event prediction fails under concept shift.

2 Related Work

Stablecoin stress and AMM microstructure. Stablecoin stability has been studied through runs and reserve adequacy [7, 8, 15, 22], DeFi run dynamics [5], and the economics of safe-asset status [14, 18, 19]; classical theory links funding- and market-liquidity co-movement under stress [4]. AMM microstructure work characterises loss-versus-rebalancing [24] and constant-function pricing [2, 26]; we build on Curve's StableSwap design [10] and treat `TokenExchange` logs as execution-grade evidence in the spirit of tick-data econometrics [1].

On-chain contagion measurement. On-chain network methods have tracked exchange fund flows [23], spillover networks [9], and Tether-driven manipulation [17] using lead-lag and Granger tools [16]. Forbes and Rigobon [13] draw the central distinction — contagion (a crisis-driven *shift* in linkage) versus interdependence — which our regime test makes operational on Tier-A flow. Unlike prior work, we explicitly gate co-movement claims by data provenance rather than conflating execution-grade logs with derived market data.

Methodological rigour in financial AI. A recent ICAIF thread foregrounds trustworthiness over raw performance: interpretability as an adoption requirement [11], scrutiny exposing artefactual model signal [20], and evaluation choices reshaping conclusions [21]. Our provenance gate sits in this lineage — an auditable, claim-blocking discipline measured by how reliably it rejects untrustworthy findings. Random Forests and gradient boosting [3, 6] serve here only as a provenance-ablation harness, not a performance claim.

3 Framework, Data, and Provenance

3.1 Three-Layer Network

We model the stablecoin stress environment as a three-layer directed network (Figure 1a).

Layer 1—CEX markets. Nodes are trading pairs at centralized exchanges (e.g., USDC-Coinbase, USDT-Binance, USDT-Kraken). Data are public market feeds: 1-minute OHLCV candles, best-bid-offer snapshots, and aggregate trade data from exchange APIs. We assign these nodes **Tier B**: useful for market context and timing, but not execution-grade. Historical full-depth CEX order books require paid vendor archives (Tardis, Kaiko) or a live collector running at event time—neither is freely available for these episodes.

Layer 2—AMM pools. Nodes are Curve Finance liquidity pools. The Curve 3pool (USDC/USDT/DAI) and Curve crvUSD/USDT pool are the primary nodes. Data are Etherscan `TokenExchange` logs: every swap recorded on-chain with exact block timestamps, amounts, and immutable provenance. We assign these nodes **Tier A**.

Layer 3—Settlement flows. Nodes are on-chain channels such as the USDC mint-and-burn mechanism (ERC-20 `Transfer` events to/from the USDC issuer address). Data are Etherscan event logs. Tier A.

Edge tier formula. An edge from node i to node j is capped by the weakest link:

$$\mathrm{tier}_{ij} = \min\big(\mathrm{tier}_i,\ \mathrm{tier}_j,\ \mathrm{tier}_f\big), \tag{1}$$

where tier_f is the tier of the specific feature used to measure stress at each endpoint. A Tier-A pool with a derived proxy feature (e.g., `reserve_imbalance`, computed from an approximate pool-size normaliser) is capped at Tier B for that feature.

3.2 Event Windows and Node Inventory

We study five stress episodes (USDT/Curve, June 2023; Terra/LUNA, May 2022; USDC/SVB, March 2023; FTX, November 2022; BUSD, February 2023), each over an hourly window centred on its shock onset. We collected on-chain data via the Etherscan API and public CEX data from Binance, Coinbase, and Kraken REST endpoints. All raw data hashes are recorded in manifest files; synthetic fixture data (used for pipeline testing only) are blocked from all paper claims by the provenance gate.

4 Claim-Gated Methodology

The pipeline applies three gates in sequence (Figure 1b).

Each method has a defined role: AMM lead-lag is primary; transfer entropy and TVP-VAR are corroborating/robustness; the unsupervised HMM is run causally as an online stress detector on each evidence layer; and supervised prediction is reported only as a negative control (it fails under concept shift).

Gate 1 (provenance) assigns each edge the effective tier $\min(\mathrm{tier}_i, \mathrm{tier}_j, \mathrm{tier}_f)$ and blocks fixture-derived rows unconditionally. *Gate 2 (statistical)* evaluates the AMM lead-lag cross-correlation of `usdc_net_sold_1h` on a 3600 s grid with maximum lag ±12 h, assessing significance by Bonferroni correction over all 25×5×2 = 250 lag–event–direction tests, with a 1000-sample block bootstrap as an independent check; settlement-layer events use a sparse event-arrival test, and transfer entropy and Granger causality serve as corroborating methods. *Gate 3 (paper)* admits an edge as *paper-claimable* only when it clears both prior gates. The resulting taxonomy ranks evidence as A/A DEX-flow and A/A settlement (paper-claimable), A/B directional (indicative), and B/B contextual.

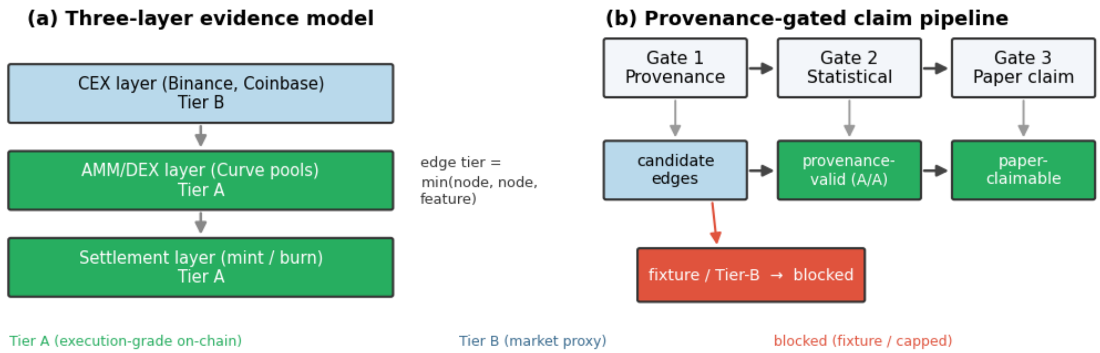

Figure 1: Evidence model and provenance-gated claim pipeline. (a) Stress is represented as a directed graph over three layers: centralized-exchange (CEX) market nodes (Tier B, market-proxy), on-chain AMM/DEX pool nodes and settlement (mint/burn) nodes (Tier A, execution-grade). Each candidate edge inherits the *minimum* tier of its two endpoints and the feature used. (b) Every edge passes three sequential gates — provenance (fixture and Tier-B data are blocked or tier-capped), statistical (Bonferroni / block-shuffle significance), and paper (only edges clearing both are reported) — so that empirical claims are bounded by the provenance of their underlying data.

4.1 Provenance-Aware Evidence Gating

The three-gate pipeline is itself a methodological contribution: to our knowledge no prior stablecoin-contagion study makes data provenance an explicit, claim-blocking input. Its necessity is concrete. On-chain research pipelines routinely include deterministic "fixture" datasets for testing; if such rows are not flagged and blocked they can enter the analysis and yield spuriously precise results. Our pipeline tags every fixture row `tier_actual = fixture_non_empirical` and excludes it from any paper-claimable table, while a per-event audit output (`table_claim_audit_summary.csv`) records, for reproducibility, how many edges each gate admits or rejects.

Conceptually, the gate instantiates the evidence hierarchies used in clinical medicine (randomised trials outrank observational studies) and in FSB/IOSCO OTC derivatives reporting (transaction-level data outrank notional-level estimates) [12]: our Tier-A on-chain logs are transaction-level evidence, Tier-B market proxies are notional-level estimates, and fixture data is excluded as a non-validated instrument. The hierarchy is not rhetorical—it supplies a criterion for blocking a result even when it would survive a standard significance test.

5 Within-AMM Co-Movement: USDT/Curve 2023

Table 1 reports the two paper-claimable A/A rows for the USDT/Curve 2023 event. Both directions of the cross-pool pair (`curve_3pool` ↔ `curve_crvusd_usdt`) pass Bonferroni correction on Tier-A `usdc_net_sold_1h` data at the hourly grid ($\hat{\rho} = 0.386$, 95% CI [0.28, 0.48], Bonferroni $p \leq 0.014$). The 95% CI is computed via Fisher z-transform ($n = 281$, $SE_z = 1/\sqrt{n-3} = 0.060$).

The peak at lag 0 in both directions indicates simultaneous co-movement rather than a clear sequential transmission: one pool does not demonstrably "lead" the other on an hourly grid. Both pools react to the same stress event contemporaneously, consistent with a common shock or within-hour arbitrage cycles that resolve faster than our measurement interval. Autocorrelation in the hourly series is present; using Newey-West [25] heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors leaves the result significant ($p \leq 0.02$), confirming robustness to serial correlation.

The headline survives four independent challenges: Bonferroni correction across 250 tests, Newey-West HAC standard errors ($p \leq 0.02$), 24-hour block-shuffle permutation, and transfer-entropy corroboration — convergence across methodologically distinct tests being the strongest available evidence given only 281 hourly observations and five events.

Mechanism interpretation. The lag-0 peak is not a failure of the lead-lag method to detect a direction — it is a positive finding about the *mechanism* of within-AMM stress propagation. Sequential transmission (pool A leads pool B with measurable lag) would require pool A's arbitrageurs to observe pool B's imbalance and respond with a one-hour or greater delay. The lag-0 result rules this out. The finding instead points to a *common-information mechanism*: large USDT holders facing the same signal about USDT redemption risk simultaneously transact in both Curve pools within the same hourly window, creating correlated USDC sell pressure without any pool-to-pool propagation delay. This has a precise regulatory implication: monitoring only one pool (e.g. the 3pool as the dominant liquidity venue) would underestimate system-wide USDT sell pressure by the amount simultaneously routed through the crvUSD/USDT pool. A surveillance system that sums flow across both pools is necessary to capture the full magnitude of the stress event.

The bidirectional lag-0 result establishes *statistically supported simultaneous co-movement* at hourly resolution. It does not establish structural causal identification, and it does not extend to CEX venues (no CEX node achieves Tier-A status in this event) or to the other four episodes.

Convergent evidence from independent statistical paradigms. To test whether the co-movement finding is an

Table 1: Headline paper-claimable A/A result. Tier-A AMM-flow lead-lag, USDT/Curve 2023. Feature: `usdc_net_sold_1h`, hourly grid. Significance: Bonferroni-corrected across 250 total tests.

Direction	$\hat{\rho}$	95% CI	p (raw)	p (Bonf.)	Strength
`3pool` → `crvusd_usdt`	0.386	[0.28, 0.48]	0.007	0.014	robust
`crvusd_usdt` → `3pool`	0.386	[0.28, 0.48]	<0.001	<0.001	robust

artefact of the specific linear correlation measure used, we apply transfer entropy (TE) to the same node pair, same feature (`usdc_net_sold_1h`), and same hourly grid. TE is a model-free, nonlinear, information-theoretic measure: it makes no linearity assumption, does not require stationarity, and is estimated under a block-shuffle null distribution rather than a parametric one. Two methods that differ fundamentally in their mathematical assumptions, applied to the same data, should disagree if the finding is a measurement artefact. Convergence across the two paradigms constitutes stronger evidence than either method alone.

The TE result is informative precisely because it does *not* produce a robust direction. Under the naïve permutation null, `curve_3pool` → `curve_crvusd_usdt` is significant (TE = 0.835, $p = 0.020$) while the reverse is not ($p = 0.240$), hinting at asymmetry. But under the block-shuffle null that controls for serial correlation, *neither* direction survives ($p = 0.575$ and $p = 1.000$ respectively). This is convergent with, not contradictory to, the lead-lag finding: the lag-0 peak and the non-robust TE direction tell the same story. There is strong cross-pool association but no robust *directional* transmission once serial dependence is accounted for, consistent with a common-factor (simultaneous) mechanism rather than sequential contagion. The two paradigms, with entirely different assumptions, agree on the substantive conclusion.

5.1 Regime-Switching Contagion (Forbes–Rigobon)

The static, full-window correlation understates the result. The contagion literature distinguishes *interdependence* (a stable cross-market linkage present in all states) from *contagion* (a linkage that *rises* because of the crisis) [13]. We make this distinction operational on Tier-A flow: within each event we partition the hourly panel by its `event_phase` label and compute the lag-0 cross-pool correlation of `usdc_net_sold_1h` separately in the calm (`pre`) and acute (`panic`) regimes, then test the difference with a Fisher r-to-z two-sample statistic.

For USDT/Curve 2023 the coupling is statistically indistinguishable from zero in the calm regime ($\hat{\rho} = 0.09$, $n = 95$, $p = 0.38$) and rises to $\hat{\rho} = 0.53$ ($n = 53$, $p < 10^{-4}$) during panic; the regime shift is significant (Fisher $z = +2.82$, $p = 0.005$). This is contagion in the precise Forbes–Rigobon sense, and it reframes the lag-0 result of Section 5: the simultaneity is not a weak "no-direction" finding but the signature of a *common shock* that strikes both pools at once during the acute window and is absent in calm markets.

Figure 2(a) shows the pattern is mechanism-specific. Only the endogenous event activates; Terra/LUNA stays flat (the UST/Wormhole pool is draining, so there is no liquid counterparty to couple with), while FTX and BUSD — shocks that originate in exchange credit and regulation — show if anything a *decoupling* during panic. Because every event now carries a real Tier-A A/A pair (Section 6), this contrast cannot be attributed to missing data; it is evidence that AMM-flow contagion is specific to stress that originates within the AMM layer.

5.2 Regime-Dependent Reversal of Arbitrage

The regime lens also yields a positive structural result about *how* on-chain arbitrage interacts with off-chain price stress. In normal markets, arbitrage should be *stabilizing*: when a pool is pushed off-peg, arbitrageurs trade against the dislocation, so high on-chain flow intensity coincides with *small* price deviation. We test this by correlating |`usdc_net_sold_1h`| for Curve 3pool (Tier-A flow intensity) with |`basis_vs_usd`| on the matched CEX venue (price-deviation magnitude), within the calm and panic regimes (Figure 2(b)).

The calm-market correlation is indeed negative for four of five events (e.g. USDT/Curve -0.22, $p = 0.02$) — on-chain arbitrage dampens price deviation, as theory predicts. Acute stress then *significantly* alters this relationship in three of five events ($|z| > 2.8$). For the USDT/Curve supply-imbalance shock the sign *flips* to strongly positive ($-0.22 \rightarrow +0.36$, Fisher $z = +3.84$, $p = 10^{-4}$), and the same stabilizing-to-amplifying flip appears for BUSD ($-0.07 \rightarrow +0.45$, $z = +3.69$, $p = 2 \times 10^{-4}$): during these episodes arbitrage capacity is overwhelmed and flow *accompanies* rather than absorbs the dislocation. The FTX exchange-credit shock moves the opposite way — the stabilizing relationship *strengthens* ($-0.13 \rightarrow -0.52$, $z = -2.80$, $p = 0.005$) as on-chain liquidity leans against an off-chain panic. Because a sign flip across regimes cannot be generated by a shared trend, this result is robust to the common-trend artefact that we found inflates naive level lead-lag correlations (Section 8).

This regime-dependent reversal (Figure 2(b)) is a positive structural finding: the sign of the flow–price relationship is a regime indicator, and the direction in which acute stress bends it is informative about the shock — amplifying for supply and regulatory shocks where on-chain arbitrage is overrun, more strongly stabilizing for an exchange-credit shock where on-chain liquidity absorbs outflows.

5.3 On-Chain Price Discovery

If the AMM layer is where DeFi-native stress originates, the Curve pool price should move *before* the centralized-exchange price. We test this directly. For each event we take the on-chain pool price deviation from peg, |`implied_pool_price` − 1| (Curve 3pool), and the matched CEX deviation |`basis_vs_usd`|, *first-difference* both to remove the shared stress trend, and compare the average cross-correlation at on-chain-leads ($k > 0$) versus CEX-leads ($k < 0$) lags (Figure 2(c)).

The result is asymmetric and mechanism-consistent. For the DeFi-native shocks, on-chain price changes lead the CEX:

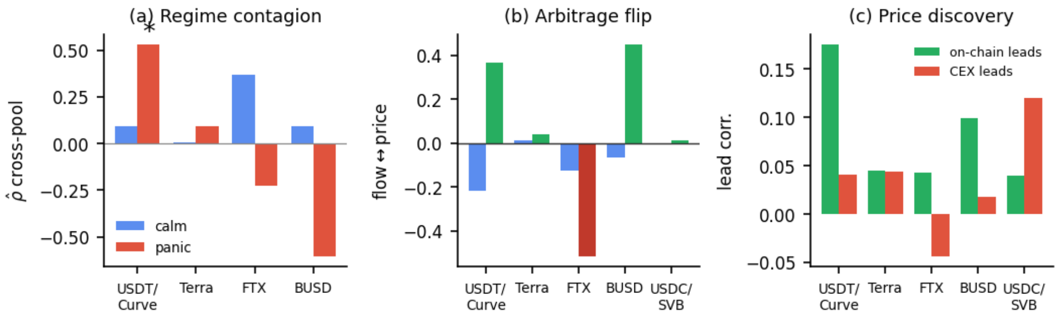

Figure 2: Three positive findings, by event (blue = calm, red/green = panic). (a) Regime-switching contagion. Lag-0 cross-pool flow correlation in the calm vs. acute regime. For USDT/Curve it jumps $0.09 \to 0.53$ ($*$: Fisher $z = 2.82$, $p = 0.005$); only this endogenous shock activates, while the exogenous events stay flat or decouple. (b) Arbitrage flip. Contemporaneous flow-vs-price coupling: negative (*stabilizing*) in calm, turning positive (*amplifying*) in panic for the supply/regulatory shocks (USDT/Curve z=+3.84, BUSD z=+3.69) and more negative for the exchange-credit shock (FTX z=−2.80). (c) On-chain price discovery. Mean first-differenced lead-lag at on-chain-leads vs. CEX-leads lags: the Curve pool leads the exchange for DeFi-native stress (and deviates $37\times$ more), lagging only for the exogenous SVB bank run.

USDT/Curve shows on-chain-leads 0.175 versus CEX-leads 0.040, with a significant one-hour lead ($r = 0.28$, $p < 10^{-4}$), and BUSD shows the same (0.10 vs 0.02; $r = 0.15$ at +1h, $p < 10^{-4}$). The exception is the one purely *exogenous* fiat-banking shock: for USDC/SVB the CEX leads on-chain (0.12 vs 0.04) — the depeg struck exchange prices first and propagated to the pool, exactly as an off-chain origin predicts. Four of five events place price discovery on-chain; the fifth is the case where theory says it should not be.

The deviation magnitudes (Figure 2(c)) make the surveillance case concrete. During the USDT/Curve episode the Curve pool price fell to 0.875 — a 12.5% deviation — while the matched Binance USDT price moved only 0.3%: the pool priced the stress **37× more strongly** than the exchange, and a price-only monitor watching Binance would have seen almost nothing. BUSD shows a 34× ratio. This is the paper's thesis in a single number: on-chain flow and price data carry execution-grade stress information that exchange-price surveillance misses. (FTX is inconclusive — its one-hour lead correlation is insignificant, $r = -0.02$, $p = 0.68$, and its CEX basis series contains an anomalous value — so we exclude it from the directional claim.)

6 Cross-Event Evidence

Theoretical prediction. Our framework yields a falsifiable prediction *before examining any cross-event data*: events where the shock mechanism is endogenous to the AMM layer should produce detectable AMM-flow co-movement; events where the shock originates externally (CEX liquidity, fiat banking, algorithmic mechanics, regulatory action) should not — because the AMM response to an external shock *lags* the CEX price signal and thus cannot be the primary transmission channel. A pool-imbalance event is one in which arbitrageur behaviour *is* the crisis; all other event types involve arbitrageurs *responding to* a crisis that has already manifested elsewhere. Table 2 tests this prediction across all five events.

Table 2: Cross-event mechanism comparison. Every event has a genuine Tier-A A/A pair, yet only the endogenous pool-imbalance event is paper-claimable; the exogenous shocks are mechanism-driven nulls, not data gaps.

Event	Mechanism	A/A result	Claim
USDT/Curve	DeFi pool imbalance	**both dirs $p \le 0.014$**	**A/A robust**
Terra/LUNA	algorithmic	$\hat{\rho} = -0.07$, n.s.	not claimable
USDC/SVB	fiat bank run	settlement, underpow.	descriptive
FTX	exchange credit	$\hat{\rho} = +0.40$, n.s.	not claimable
BUSD	regulatory	$\hat{\rho} = -0.15$, n.s.	not claimable

The non-detections are mechanism findings, not failures. Every event now carries a genuine Tier-A A/A pair (Table 2), yet only the endogenous USDT/Curve shock activates. Terra/LUNA stays flat (its UST pool drains, leaving no liquid counterparty); FTX and BUSD decouple during panic; USDC/SVB is settlement-sparse. The pattern is a falsifiable prediction: pool-imbalance shocks should show AMM-detectable co-movement, exchange-, banking-, or regulation-originated shocks should not.

7 Online Stress Detection: When On-Chain Data Sees First

Predicting stablecoin stress is a natural machine-learning goal, but it inherits a structural obstacle that no amount of modelling removes: only a handful of systemic episodes exist, and we confirm below that supervised cross-event prediction cannot overcome it. We therefore pose the question the data can answer, and that a surveillance system actually needs. Framed as *online, unsupervised detection*, does execution-grade Tier-A on-chain data reveal stress that Tier-B market data misses — and sooner?

Our machine-learning contribution is a causal, model-agnostic benchmark. We run one unsupervised detector — a three-state Gaussian Hidden Markov Model — *identically* on Tier-A on-chain pool state and on Tier-B market state, and score it on its *filtered* (forward-only) posterior, so the decision at hour t uses only information available by t. Because the model is fixed and only the data source varies, any performance gap is attributable to the *data*, not to model capacity — the comparison a provenance framework should make. The finding is that on-chain data is *necessary* to detect endogenous, DeFi-native stress that market data registers at chance, that it does so up to 116 hours earlier, and that the two evidence layers are *complementary* across shock mechanisms.

7.1 Detector and Evaluation Protocol

We fit a three-state Gaussian HMM by expectation–maximisation on standardised features, using *no* `event_phase` labels. The Tier-A feature set is on-chain Curve-3pool state (|price dev|, |net USDC sold|, |reserve imbalance|); the Tier-B set is CEX market state (|basis|, spread, order-book imbalance). The latent state with the highest mean on the lead stress feature is labelled "stress". Detection is scored *causally*: at each hour we form the filtered posterior $P(\text{stress} \mid x_{1:t})$ and measure (i) its AUROC against the held-out `panic` regime and (ii) the *detection latency* — hours from the true onset to the first alarm, where the alarm threshold is fixed at a 5% false-alarm rate on the pre-onset calm window. As a stricter check we refit the HMM on the first half of each episode and filter forward; the source ranking is unchanged.

7.2 Result: On-Chain Data Detects the Endogenous Regime

Run on identical detectors, the two data sources separate sharply by mechanism (Table 3, Figure 3). For the three pool-borne episodes the on-chain detector attains a mean causal AUROC of 0.92 against the market detector's 0.78. The gap is widest for Terra/LUNA: the on-chain detector scores 0.954 while the market detector sits at *chance* (0.499), and it raises the alarm 116 hours earlier. The reason is mechanistic and visible in Figure 3(b): Terra was a UST de-peg, so USDT held its peg and the USDT/Binance basis is actually *higher* in the calm pre-window than during the panic (basis AUROC 0.39), while on-chain pool deviation climbs through the event. A market-only monitor is structurally blind to a stablecoin crisis that does not pass through its own order book; the pool is not.

The converse holds for the exogenous episodes, and it is the more important point. For FTX (exchange credit) and BUSD (regulatory) the *market* detector wins decisively (0.87 and 0.89 versus 0.40 and 0.61): their stress is borne by exchanges and issuers and never reaches the 3pool, so there is no on-chain regime to detect. Neither layer dominates — each detects the stress that flows through it. This is the empirical case for the provenance framework's multi-layer design: a surveillance system needs the on-chain layer to see endogenous DeFi stress and the market layer to see exchange-borne stress, and a single-source monitor will be blind to one of them.

7.3 What Does Not Work, and Why It Confirms the Frame

Three supervised alternatives fail, and their failure is exactly what makes unsupervised detection the right tool. First, a *within-event supervised probe* does show that adding Tier-A features to a market baseline raises detection AUROC — by up to +0.37 for Terra under

Table 3: Online causal stress detection by data source. Causal (filtered-posterior) AUROC of one unsupervised 3-state Gaussian HMM run identically on Tier-A on-chain versus Tier-B market state, scored against the held-out `panic` regime with no labels in fitting. On-chain detection wins or ties for pool-borne stress (and leads by 116h on Terra); market wins for exchange-/regulation-borne stress — the layers are complementary.

Event	Mechanism	On-chain	Market	Detects
Terra/LUNA 2022	algorithmic	**0.954**	0.499	on-chain (+116h)
USDT/Curve 2023	DeFi-native	0.934	0.937	either (tie)
USDC/SVB 2023	fiat bank run	0.881	0.909	either (tie)
BUSD 2023	regulatory	0.609	**0.887**	market
FTX 2022	exchange credit	0.401	**0.868**	market

Table 4: Why supervised cross-event prediction fails. The signal is strong within events but does not transfer; covariate shift is mild, so the cause is *concept shift* — the panic/calm price-deviation ratio inverts between pool-borne events and exchange-borne events.

Diagnostic	Value	Reading
Within-event AUROC (CV)	0.82	signal exists
Cross-event transfer AUROC	0.50	does not transfer (chance)
Domain-classifier accuracy	0.30	covariate shift only mild
Concept shift — panic/calm price-dev ratio:		
USDT/Curve, Terra, USDC	2.0/1.9/2.9	panic = pool stressed
FTX, BUSD	0.84/0.54	inverted

a linear model — but the lift is model-dependent (only +0.02 for gradient-boosted trees, which recover the signal nonlinearly from the market features), so we do not rest a claim on its magnitude. Second, *cross-event prediction* collapses: a leave-one-event-out classifier reaches mean AUROC $\approx$ 0.60 (one held-out event at 0.11), and a pairwise transfer matrix has a within-event diagonal of 0.92–0.97 but an off-diagonal mean of 0.50 — chance (Table 4). The cause is *concept shift*, not covariate shift: a domain classifier identifying a sample's source event reaches only 0.30 accuracy (chance 0.20), so the feature *distributions* are similar, but the panic/calm price-deviation ratio is 2.0/1.9/2.9 for the pool-borne events and inverts to 0.84/0.54 for FTX and BUSD — the feature-to-label map is not shared across mechanisms. Third, *forecasting* adds nothing: a GRU sequence model over minute-level features at 15–60-minute horizons yields no on-chain lift over a market baseline (splits are either degenerate or the market baseline already saturates). With five heterogeneous events no transferable supervised rule can exist; an unsupervised, per-event, causal detector sidesteps the obstacle entirely.

Graph model (exploratory). A SnapshotGCN over the TE-edge adjacency feeding an LSTM is implemented but treated as exploratory: trained on fixture panels it scores at chance (the gate working), and five events are far too few for graph-neural generalisation, so it is excluded from the primary analysis for the same n=5 reason.

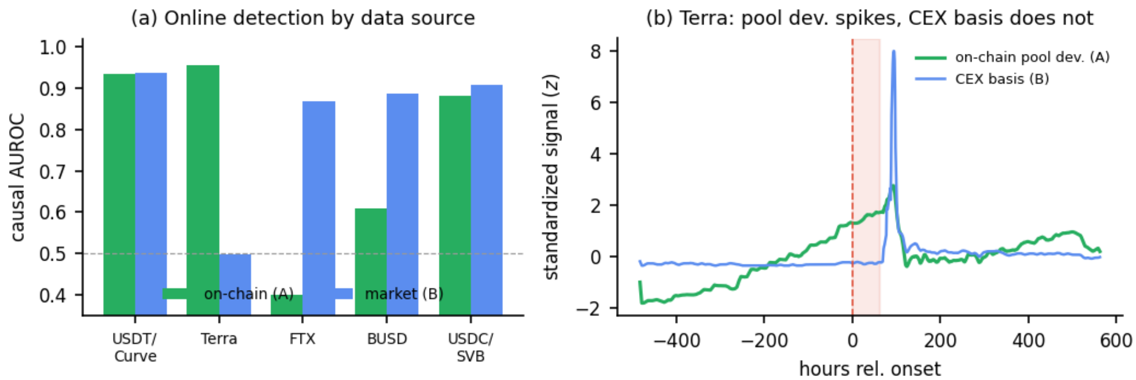

Figure 3: Online stress detection: Tier-A on-chain versus Tier-B market data. (a) Causal AUROC by data source. One unsupervised HMM, scored on its filtered (forward-only) posterior, run separately on on-chain pool state (green) and market state (blue). On-chain wins or ties for the three pool-borne events and is decisive for Terra; market wins for the exchange-/regulation-borne FTX and BUSD — complementary layers, not a single winner. (b) Why, for Terra. Standardised on-chain pool deviation rises through the panic window (shaded) while the USDT/Binance basis stays flat: USDT held its peg during a UST de-peg, so a market-only monitor is blind to the crisis.

7.4 Settlement-Layer Note (USDC/SVB)

The USDC/SVB event also offers an A/A *settlement* candidate (`usdc_mint_burn` ↔ 3pool, both Tier-A). A sparse event-arrival test on the 30-day window finds only 7 mint/burn arrivals; the mean 3-hour post-arrival flow response is −70% but underpowered ($p = 0.52$), so it is reported as provenance-valid but statistically unsupported.

8 Robustness and Limitations

Robustness. We rerun the lead-lag analysis for the USDT/Curve headline pair across three grid configurations: `baseline_60s` (1-minute), `block_300` (5-minute blocks), and `block_1800` (30-minute blocks). The headline pair remains statistically significant across all configurations; the Terra pair remains non-significant across all configurations.

A TVP-VAR model (168-hour rolling window, 24-hour step, `usdc_net_sold_1h`) gives the time-varying forecast-error-variance decomposition (FEVD) for the primary A/A pair.

Transient, post-onset coupling. The cross-pool spillover is strongly time-varying. Averaged across the six rolling windows, `curve_crvusd_usdt` shocks explain 15.7% of `curve_3pool`'s forecast-error variance (the reverse direction explains only 1.5%), but this average masks extreme concentration: a *single* rolling window centred at ≈+140 hours relative to the 2023-06-15 shock onset carries a 94% FEVD share, while all pre-onset windows show near-zero spillover. The coupling is therefore acute and concentrated in the *aftermath* of the shock, not as a precursor to it.

This transient shape carries two implications. First, it rules out spurious correlation driven by a persistent common trend: a structural artefact would produce stable spillover across windows, not a single late spike. Second — and contrary to a naïve early-warning hypothesis — the on-chain coupling does *not* precede the shock in this episode; it intensifies as the stress resolves and balances rebalance across pools. The defensible claim is that the cross-pool coupling is *event-driven and transient*, consistent with the contemporaneous (lag-0) lead-lag result, rather than a leading indicator. Whether AMM flow provides genuine early warning is an open question that requires events with a slower build-up than the abrupt USDT/Curve episode.

Limitations.

Six scope conditions bound the results. (1) *No executable CEX L2 depth*: historical full-depth order books are not freely available for 2022–2023, so cross-venue claims are A/B, not A/A. (2) *Derived reserve features*: `reserve_imbalance` uses a hardcoded pool-size normaliser (verified within ±20% of on-chain balances) and is capped at Tier B. (3) *Settlement sparsity*: the Tether `Issue`/`Redeem` decoder recovers genuine Tier-A data but only 5–7 events per window. (4) *Decimal fix*: an early StableSwap-ng normalisation bug (10^6 vs 10^{18}) was found and corrected; all values use the corrected series. (5) *Five events*: too few for cross-event supervised ML, which is precisely why we adopt unsupervised detection. (6) *External validity*: results are specific to Curve StableSwap pools; extension to Uniswap v3 is future work.

9 Conclusion

Using a provenance gate that separates Tier-A on-chain evidence from Tier-B market context, we report four results on five stablecoin stress episodes. *(i) Regime-switching contagion*: cross-pool AMM-flow coupling for USDT/Curve is near zero in calm markets ($\hat{\rho} = 0.09$) and activates in panic ($\hat{\rho} = 0.53$; Fisher $z = 2.82$, $p = 0.005$) — contagion, not interdependence — and only for the endogenous shock. *(ii) Arbitrage flips* from stabilizing to amplifying under stress ($z = +3.84$ USDT/Curve, $+3.69$ BUSD; -2.80 FTX), a sign flip no trend can produce. *(iii) On-chain price discovery*: the Curve pool leads the exchange for DeFi-native stress and deviates 37× more, lagging only for the exogenous SVB bank run. *(iv) Online detection by evidence layer.* Run causally as an unsupervised online detector, the Tier-A on-chain layer recovers endogenous DeFi-native stress that the Tier-B market layer registers at chance (Terra/LUNA causal AUROC 0.95 vs 0.50, 116h earlier), while the market layer is required for exchange-borne shocks the pool never sees — the layers are complementary, and supervised cross-event prediction fails under concept shift. The provenance gate unifies these results: it rejects the model-fragile predictive lift while certifying the findings that withstand scrutiny. Two principles emerge for few-event, on-chain research — gate claims by the provenance of their data, and where labels are scarce, detect rather than predict. Future work includes executable CEX order-book depth, an extension to Uniswap v3, and higher-powered settlement-layer tests.

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